Sprint 1 Handover Notes

Config-Driven RL for Just-in-Time Adaptive Interventions

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1 Problem Definition

Physical inactivity is a leading global health risk. Just-in-Time Adaptive Interventions (JITAIs) use mobile/wearable devices to deliver personalised behaviour-change prompts at moments of opportunity [1]. Designing effective JITAI policies requires iteration, but real-world trials are slow and expensive.

We formulate the problem as a Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$:

- $\mathcal{S}$: 36-state factored space (step_bin $\in$ {inactive, moderate, active}, sleep $\in$ {good, poor}, day_of_week $\in$ {weekday, weekend}, burden $\in$ {low, medium, high})
- $\mathcal{A}$: 4 actions (idle, movement_suggestion, goal_reminder, journal)
- $\mathcal{R} = \alpha \cdot f(\text{step\_bin}) + (1-\alpha) \cdot g(\text{sleep}) - \lambda \cdot \mathbb{I}[\text{action} \neq \text{idle}]$
- 5 decision points per day, 90-day default episode, 50 seeds per agent

2 Existing Work

HeartSteps V1/V2 (Klasnja et al., 2019 [1]; Liao et al., 2019 [2]) — First RL-enabled MRT for physical activity. V1 ATE +35 steps (p=0.06). V2: Thompson Sampling with dosage tracking and proxy value function; 78.4% of participants improved, mean +29.75.

PEARL (Lee et al., 2025 [3]) — Largest RL-for-PA RCT to date (n=13,463, 4 arms). RL bandit with COM-B informed 155-nudge bank. RL +296 steps vs control (p=0.0002) at 1 month; sustained +210 at 2 months. Provides the real-world effect-size calibration target.

StepCountJITAI (Karine & Marlin, 2024 [4]) — OpenAI Gym-style simulation environment for PA interventions, DQN ≈3,000 return.

Bandit theory Russo & Van Roy (2018) establish TS regret bound $O(\sqrt{dT \log T})$ for linear contextual bandits. Trella et al. (2022) provide design principles for RL in digital health.

LLM-based simulation Park et al. (2023) demonstrate generative agent simulations of human behaviour. Wang et al. (2024) use LLMs to simulate patient responses (Patient-Ψ). Lu et al. (2025) benchmark LLM agents for multi-turn human behaviour simulation.

Data-Driven Simulators & Burden Wang et al. (2021) [5] developed data-driven behavioral simulators incorporating frequency constraints to explicitly mitigate user over-notification and habituation, maximizing long-term intervention responsiveness.

User Heterogeneity & Pooling Tomkins et al. (2021) [6] introduced *IntelligentPooling* to dynamically balance personalization with group-level data sharing. Similarly, Ghosh et al. (2024) [7] proposed *reBandit*, utilizing random effects and Empirical Bayes to optimize online hyperparameter adjustment in highly noisy mHealth environments.

Count-Based Bandits Recent work extending Thompson Sampling to non-linear count data models [8] (e.g., Poisson and Negative Binomial regressions) demonstrates significant reductions in variance and regret when modeling highly skewed proximal health outcomes compared to standard linear baselines.

Power-Constrained Exploration Yao et al. (2021) [9] established meta-algorithms for probability clipping ($\pi_{min} \leq P(A|S) \leq \pi_{max}$) to protect exploration, ensuring that deploying adaptive personalization policies does not compromise the post-trial statistical power required to test overall intervention efficacy.

3 Our Work: Config-Driven Environment

The framework is driven entirely by YAML configuration files validated through Pydantic schemas. No source code changes are required to define new MDPs, agents, or experiments. Core components:

- **State:** Factored state with cyclic advances (day_of_week) and rolling window counts (burden tracking over last N steps). Day-boundary sleep routing.
- **Transitions:** Three types — rule_based (hand-curated), random (Dirichlet tables, baseline), bootstrap (LLM-generated JSON tables).
- **Rewards:** Safe arithmetic expression parser (AST-based). Variables from state factors or action, with configurable mappings.
- **Agents:** 7 implemented — fixed, random, epsilon-greedy, decaying epsilon-greedy, Thompson Sampling, UCB, HeartSteps V2. All support contextual variants.
- **Evaluation:** Multi-seed benchmark runner, regression test suite (MVP, sprint1_random, sprint1_bootstrap), comparison table formatting.

4 Our Work: Synthetic LLM Patients

We generate patient transition probability tables by prompting LLMs to sample plausible next-state behaviours:

Pipeline: Prompt generation → concurrent API calling (litellm, 200 workers) → JSON parsing → probability table aggregation.

- 22,320 prompts per persona (full state-action-timestep factorial, N=10 per cell)
- 5 distinct behavioural personas (base, goal_driven, social_responder, resistant, stable_maintainer)
- **Total cost: $0.27** (DeepSeek V4 Flash via OpenCode Zen)
- 42 JSON transition tables (6 within-day + 1 day-boundary per persona/initial model)
- Analysis pipeline: parse rate, output distributions, cell consistency, burden interactions, step histograms, transition matrix visualizations (85 PDF figures)
- 36-paper literature review across 8 dimensions

5 Initial Results

HeartSteps V2 achieves **97–100% of the optimal DP bound** across all 7 persona/initial-model configurations, dominating bandit baselines by 10–40 percentage points:

Persona	Optimal DP	HeartSteps (% opt)	Best Bandit (% opt)
Stable Maintainer	441.8	440.8 (99.8%)	402.5 (91.1%)
Base	379.7	379.1 (99.9%)	327.8 (86.3%)
Goal-driven	414.1	414.5 (100.1%)	323.0 (78.0%)
Social Responder	313.5	306.8 (97.9%)	140.1 (44.7%)
Initial DeepSeek	381.6	373.6 (97.9%)	297.5 (78.0%)
Initial GLM 5.2	374.6	374.0 (99.8%)	326.8 (87.2%)
Resistant	214.6	163.5 (76.2%)	151.7 (70.7%)

Key findings: (1) Large persona effects — Stable Maintainer (441) vs Resistant (215). (2) Epsilon-greedy is the best bandit but still 10–40pp behind HeartSteps. (3) Time-of-day features and step activity levels are most predictive. (4) HeartSteps selects idle ≈62% of the time — personalisation is partly about knowing when *not* to intervene.

6 What is Next

1. **Sensor Foundation Models.** Wearable FM POC (2,754 lines): NormWear (768-dim embeddings), FM-guided action space (15 interventions, 3 archetypes), Thompson Sampling with FM context keys.
2. **Other LLMs.** Extend bootstrapping to GLM 5.2, Qwen, Claude. Multi-turn persona dialogues for richer state transitions.
3. **Better Prompting.** COM-B behaviour-change taxonomy integration. Structured JSON schema enforcement. Chain-of-thought prompting.
4. **Synthetic Clinical Results.** Calibrate transitions to PEARL effect sizes (+200–300 steps). Integrate All of Us (n=59K) and UK Biobank (n=100K) for learned transition models.
5. **Deep RL agents.** Extend beyond bandits to DQN and policy-gradient methods for temporal credit assignment. Add power-constrained exploration [9] and count-based likelihoods [8].

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